

Chapter 1

INTRODUCTION

Counterfeit medicines have become a major global concern that threatens patient safety and healthcare quality. Fake medicines may contain incorrect ingredients, improper dosage, or harmful substances that can lead to treatment failure and serious health complications.

The objective of this project is to design and develop an AI Based Fake Medicine Detector that uses Artificial Intelligence and image processing to verify medicine authenticity. The system captures medicine images and compares packaging details with stored database records to identify counterfeit products.

The solution provides a fast, reliable, and user-friendly method for medicine verification while promoting healthcare safety.

1.1 Brief history of Fake medicines

Traditional medicine verification methods relied on manual inspection and laboratory testing. Although accurate, these approaches are time-consuming and expensive.

With advances in Artificial Intelligence and computer vision, automated medicine verification systems have emerged. Modern systems analyze packaging patterns, barcodes, and visual features to identify fake medicines quickly and efficiently.

1.2 Modern Fake Medicine Detector

- Accuracy in fake medicine detection
- Fast medicine verification
- Secure data storage
- Easy portability and usability
- Scalability for large medicine databases
- Low operational cost

Chapter 2

Problem Statement

2.1 Description

Counterfeit medicines are increasingly entering the pharmaceutical supply chain, making it difficult for consumers to identify authentic products. Existing verification methods often require expert knowledge or laboratory testing.

There is a need for an intelligent, affordable, and easy-to-use system that can verify medicine authenticity using Artificial Intelligence.

2.2 Challenge Statement

Developing the system involves challenges such as:

- Accurate image processing
- Dataset availability
- Different medicine packaging styles
- Real-time detection
- Maintaining reliability and speed

Chapter 3

3.1 Design Thinking Process

a) Empathize: Understand difficulties faced by consumers and pharmacists in identifying counterfeit medicines.

b) Define: Identify problems in current verification methods.

c) Ideate: Generate solutions using AI and image analysis.

d) Prototype: Develop detection model with camera and AI.

e) Test: Evaluate system accuracy using genuine and fake medicine samples.

3.2 Methodology

- Capture medicine image
- Extract visual information
- Process using AI model
- Compare with database
- Generate result
- Alert user

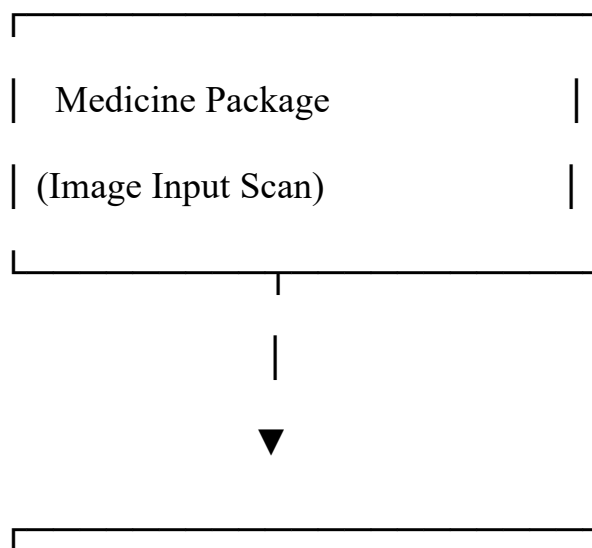
3.3 Prototype Description

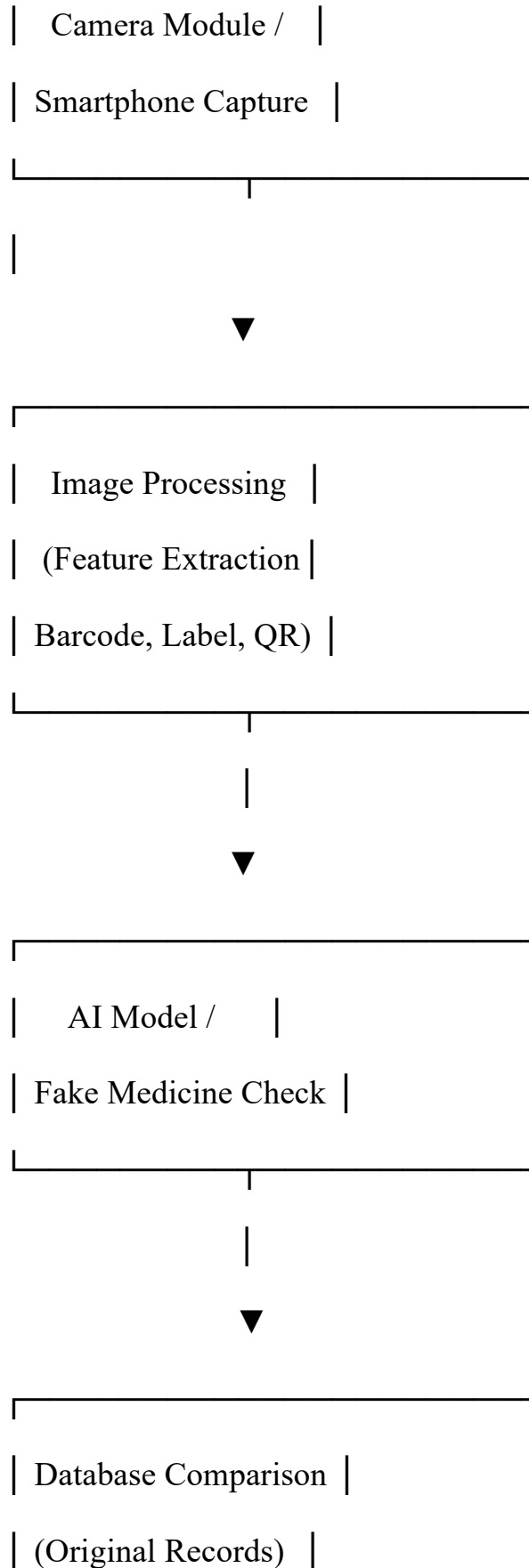
3.3.1 Materials Used

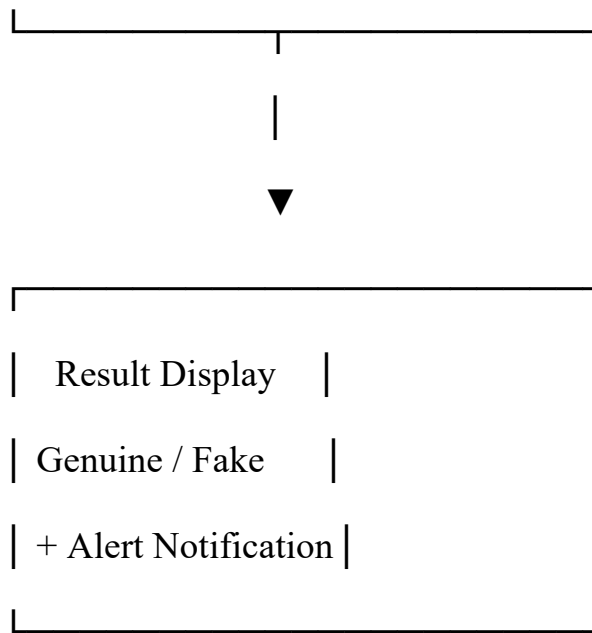
The prototype includes:

- Camera Module
- AI Detection Model
- Database
- Display Module
- Alert System

3.3.2 System Diagram







Chapter 4

Implementation

```
# app.py

from flask import Flask, render_template, request

import pytesseract

import sqlite3

import re

app = Flask(__name__)

@app.route("/")

def home():

    return render_template("home.html")

@app.route("/scanner", methods=["GET", "POST"])

def scanner():

    text = pytesseract.image_to_string(img)

    dates = re.findall(r'\d{2}/\d{4}', text)

@app.route("/assistant", methods=["GET", "POST"])
```

```
def assistant():  
  
    if "fever" in question:  
  
        answer = "Use Paracetamol"  
  
@app.route("/complaint", methods=["GET","POST"])  
  
def complaint():  
  
    conn = sqlite3.connect("complaints.db")  
  
app.run(debug=True)
```

MEDICINE DATABASE:

medicine_db.py

```
medicines = {  
  
    "DOLO 650": {  
  
        "composition": "Paracetamol",  
  
        "uses": "Fever and pain",  
  
        "side_effects": "Nausea"  
  
    },  
  
    "PARACIP-500": {  
  
        "composition": "Paracetamol",  
  
        "uses": "Pain relief",  
  
        "side_effects": "Dizziness"  
  
    }  
  
}
```

HOME HTML

```
<!-- home.html -->  
  
<h1>AI Medicine Safety Detector</h1>  
  
<a href="/scanner">Scan Medicine</a>  
  
<a href="/assistant">AI Assistant</a>  
  
<a href="/complaint">Complaint</a>
```

INDEX HTML

<!-- index.html -->

<form method="POST" enctype="multipart/form-data">

<input type="file" name="image">

<button type="submit">

Analyze Medicine

</button>

</form>

<p>{{ medicine }}</p>

<p>{{ composition }}</p>

<p>{{ uses }}</p>

AI ASSISTANT:

<!-- assistant.html -->

<form method="POST">

<input type="text" name="question">

<button>Ask</button>

</form>

<p>{{ answer }}</p>

MEDICINE INFORMATION:

<!-- medicine_info.html -->

<form method="POST">

<input type="text" name="medicine">

<button>Search</button>

</form>

<p>{{ result }}</p>

COMPLAINTS:

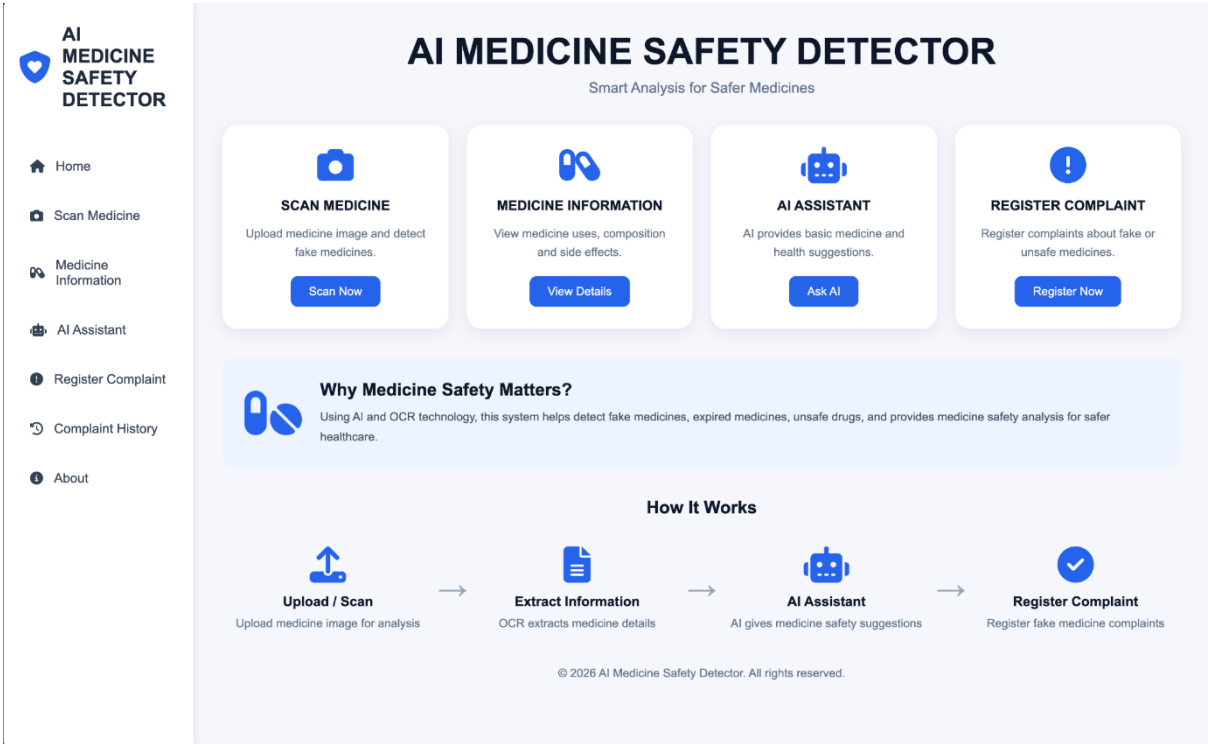
<!-- view_complaints.html -->

<h1>Complaints</h1>

{% for c in complaints %}

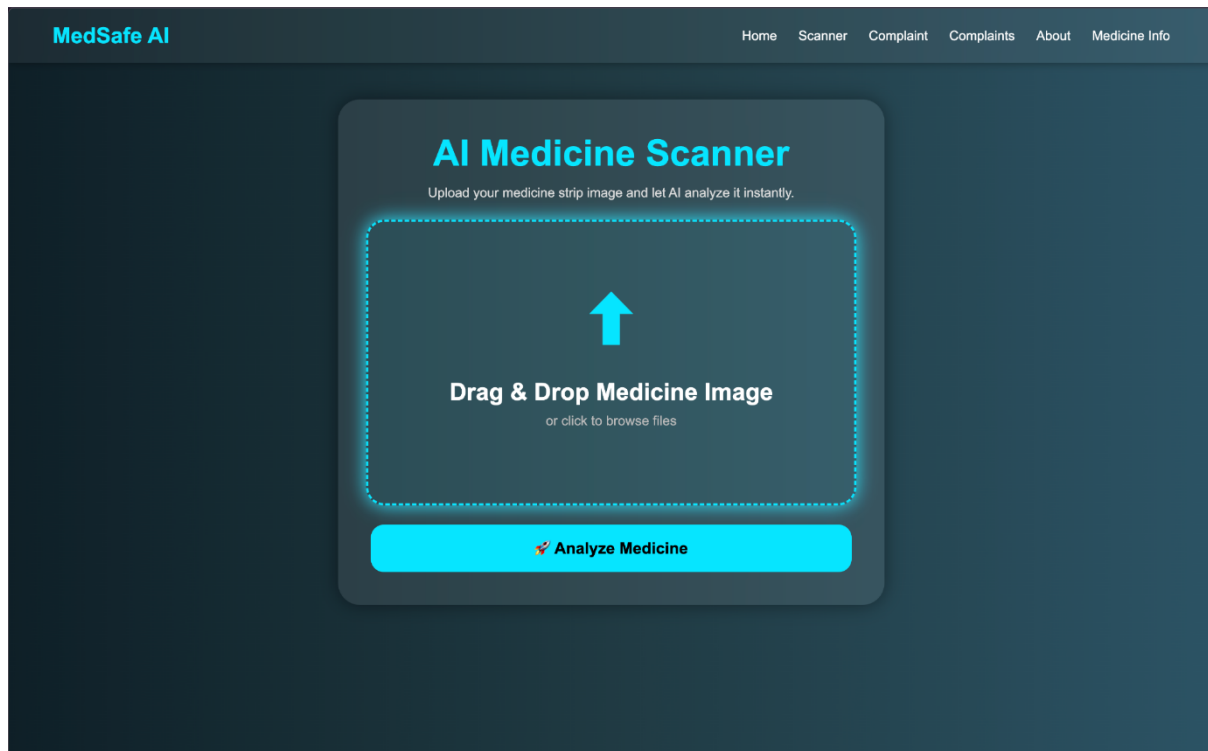
<p>{{ c }}</p>

{% endfor %}



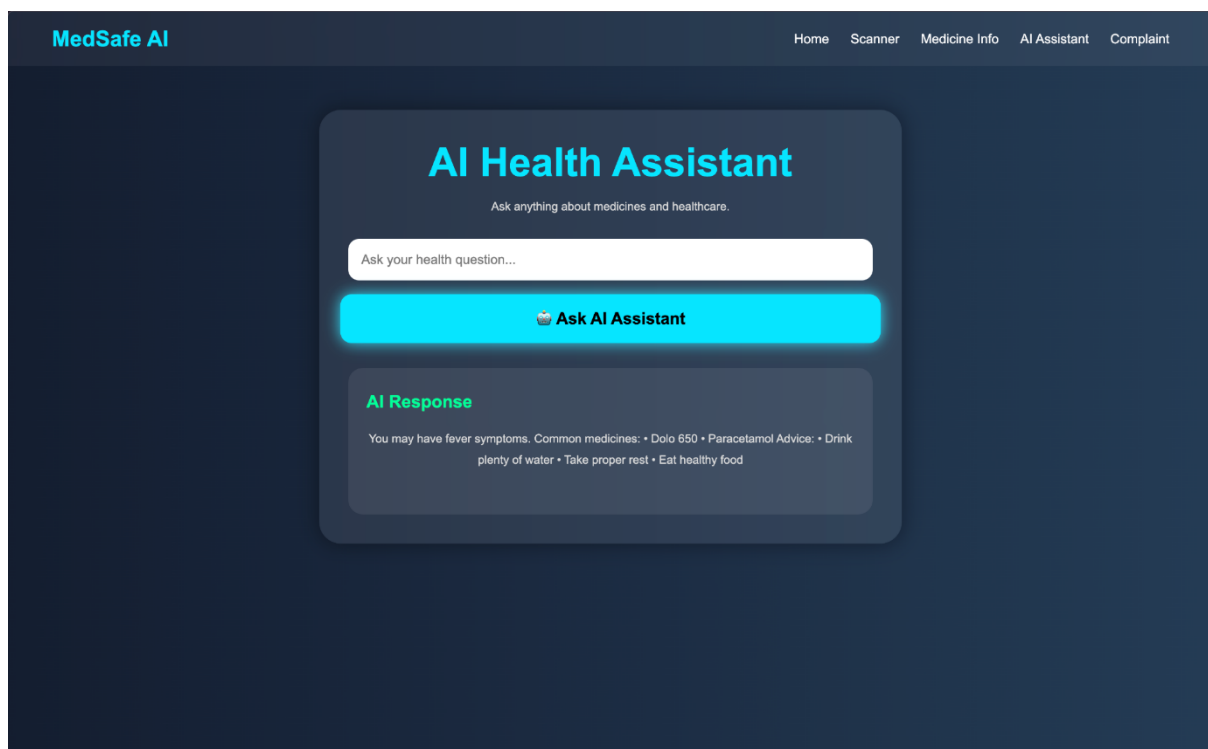
Home Dashboard

This dashboard serves as the main interface of the AI Medicine Safety Detector, providing access to medicine scanning, medicine information, AI assistance, and complaint registration features.



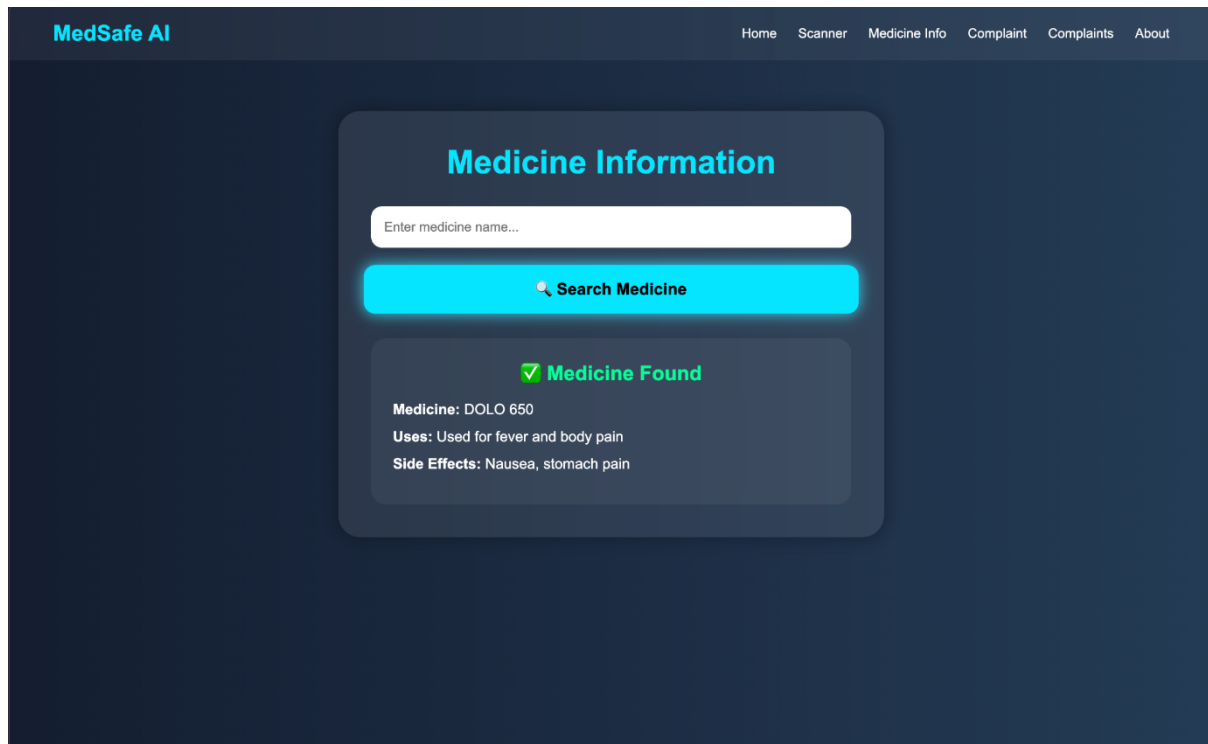
Medicine Scanner Page

This page allows users to upload or scan a medicine image, which is analyzed by the AI system to detect counterfeit or unsafe medicines.



AI Health Assistant Page

This page provides AI-generated healthcare guidance by answering medicine-related queries and offering basic health recommendations.



Medicine Information Page

This module enables users to search for a medicine and view details such as its uses, composition, and possible side effects

Chapter 5

Results and Analysis

User Testing & Feedback

Testing was conducted using medicine images.

Observations:

- Original medicine → Genuine detected

- Counterfeit medicine → Fake detected
- Damaged package → Re-scan requested

Advantages:

- Fast verification
- Improved healthcare safety
- Reduced counterfeit risks

Chapter 6

Conclusion & Future Work

The AI Based Fake Medicine Detector provides a smart and affordable solution to identify counterfeit medicines. By combining Artificial Intelligence and image processing, the system improves medicine authentication and protects users from health risks.

Future Enhancements

- QR code authentication
- Mobile application integration
- Cloud database support
- Blockchain-based tracking
- Voice-assisted verification

References

Research Papers

- <https://ieeexplore.ieee.org/>

- AI in Medicine Authentication
- Image Processing for Healthcare

Text books

- Artificial Intelligence Fundamentals
- Machine Learning Applications

Online articles

- [ibm.com 7](#)
- Medicine Safety Technologies
- AI Healthcare Solutions

Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles